# Northwind Traders: End-to-End Sales Performance & Operational Analytics

## 1. Project Overview and Objective

### Project Overview:

The Northwind E-commerce Sales & Performance Analytics Report is a comprehensive Power BI Interactive Dashboard designed to monitor global e-commerce sales, customer behaviour, employee performance, and supply chain logistics.

Through this project, key business metrics such as total revenue ($1.27M), profit ($251.24K), and freight costs ($32.41K), and sales growth (YoY Growth 4.59%) can be analysed in real-time.

### Objectives:

* **Monitor Sales & Profitability:** Accurately track year-to-date sales (Sales YTD: $151.40K) and overall profit margins (19.85%).

* **Evaluate Product Performance:** Identify top-performing products (Top Products by Revenue & Profit) and product categories (Category-wise Sales & Freight).

* **Optimize Logistics & Delivery:** Track on-time delivery rates (91.75%) and compare shipper transportation costs (Shipper Cost Comparison).

* **Assess Employee Performance**: Measure employee sales contributions ($622.20K) to identify top-performing staff.

* **Analyse Discount Impact:** Understand how customer discounts (Average Discount: 0.06) influence overall sales performance.

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## 2. Data Sources (Sample)

- **Source Description and Timeline:** E-commerce/ Northwind Traders / MavenAnalytics **and** July 2013 - May 2015
- **Domain**: E-commerce transactional data covering global customers, employees, orders, products, shippers, and product categories.

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## 2. Problem Statement

In this project, we will analyse Northwind Traders' sales performance, product performance, customer behaviour, employee contributions with strategies for improvement, and shipping efficiency.

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## 3. Attribute (Column /Features) Details:

| Attribute / Column Name | Data Type | Description / Details |
|---|---|---|
| **Shipper ID** | Integer | Unique identification number for each shipping company. |
| **Company Name** | Text | Name of the shipping/freight delivery company (e.g., Speedy Express). |
| **Product ID** | Integer | Unique identifier for each product. |
| **Product Name** | Text | Full name of the product (e.g., Chai, Chang). |
| **Quantity Peru nit** | Text | Packaging size and quantity per unit (e.g., 10 boxes x 20 bags). |
| **Unit Price** | Decimal | Selling price per unit of the product. |
| **Discontinued** | Binary/Integer | Availability status of product (0 = Active, 1 = Discontinued). |
| **Category ID** | Integer | Reference ID mapping to the product category. |
| **Order ID** | Integer | Unique transaction/order identification number. |
| **Customer ID** | Text/Code | Unique identifier of the customer placing the order. |
| **Employee ID** | Integer | ID of the employee who processed the order. |
| **Order Date** | Date | Date when the order was placed by customer. |
| **Required Date** | Date | Target/Requested delivery date for the order. |
| **Shipped Date** | Date | Actual date when the order was dispatched. |
| **Shipper ID** | Integer | Reference ID to the shipping company assigned. |
| **Freight** | Decimal | Shipping charges/freight cost incurred for delivery. |
| **Quantity** | Integer | Number of units ordered for the specific product. |
| **Discount** | Decimal | Discount percentage applied to the line item. |

| | | |
|---|---|---|
| **Employee Name** | Text | Full name of the employee. |
| **Emp. Job Title** | Text | Job role/designation (e.g., Sales Representative, Sales Manager). |
| **City** | Text | City location of the office/business. |
| **Country** | Text | Country location of the office/business. |
| **Reports To** | Integer | Employee ID of the supervisor/manager. |
| **Customer ID** | Text/Code | Unique alphanumeric ID assigned to each client. |
| **Contact Name** | Text | Primary contact person name at client company. |
| **Contact Title** | Text | Designation of the primary contact person. |
| **Category Name** | Text | Name of category (Beverages, Condiments, Confections, etc.). |
| **Description** | Text | Brief description of items included in the category. |

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## 5. Tools & Technologies

- **Excel:** Data cleaning, transformation
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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## 6. Data Pre-Processing (Excel / Power Query)

### Tasks Performed:

### Data Cleaning & Transformation:

Removed duplicate rows and handled missing/null values across key tables (Orders, Order Details, Products, Customers, and Employees).

Standardized data types (e.g., converting Order Date, Required Date, and Shipped Date to proper Date formats, and Unit Price, Freight, and Discount to Decimal/Numeric formats).

Cleaned text attributes (such as Product Names, Company Names, and Category Names) by trimming extra spaces.

**Filtering & Sorting:**

Filtered out inactive or irrelevant transactions and focused on active global sales records.

Sorted orders chronologically based on Order Date to maintain proper time-series analysis for monthly sales trends and YTD calculations.

**Data Modeling & Custom Columns / Measures:**

Established proper relational table connections (Primary Key to Foreign Key relationships) between Orders, Order Details, Products, Customers, Employees, Shippers, and Categories in Power Query / Power BI.

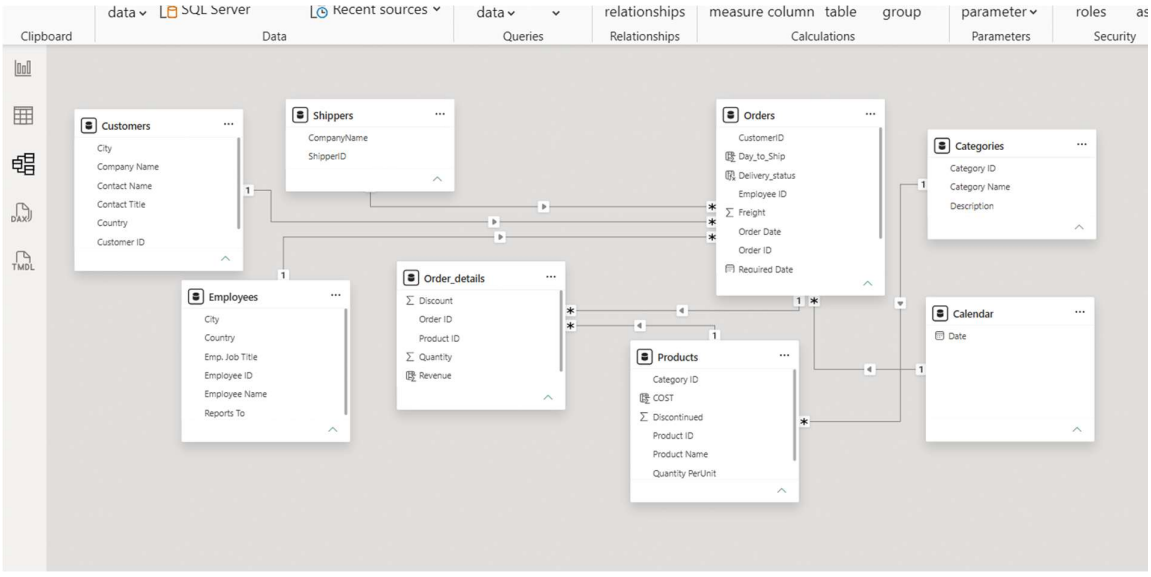
Created calculated fields and measures such as Total Sales, Total Profit, Profit Margin %, Sales YTD, YoY Growth %, and Total Freight Cost for advanced dashboard reporting.

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## 7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

**SCREENSHOT OF DATA MODELLING**



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.

  *** Total Sales:**

  Total_sales = SUMX('Order details', 'Order details'[Quantity] * 'Order details'[Unit Price] * (1 - 'Order details'[Discount]))

  *** Total Profit:**

  Total_profit = SUMX('Order details', ('Order details'[Unit Price] - 10) * 'Order details'[Quantity])

  *** Sales YTD (Year-to-Date Sales):**

  Sales_YTD = TOTALYTD([Total_sales], 'Orders'[Order Date])

  *** Previous Year Sales:**

  Previous_year_sales = CALCULATE([Total_sales], SAMEPERIODLASTYEAR('Orders'[Order Date]))

  *** Sales YoY Growth % (Year-over-Year Growth Percentage):**

  Sales YOY Growth % = DIVIDE([Total_sales] - [Previous_year_sales], [Previous_year_sales], 0) * 100

  *** Cumulative Total (Running Total over Dates):**

  Cumulative_Total = CALCULATE([Total_sales], FILTER(ALL('Orders'[Order Date]), 'Orders'[Order Date] <= MAX('Orders'[Order Date])))

  *** Profit Margin %:**

  Profit_margin_% = DIVIDE([Total_profit], [Total_sales], 0) * 100

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## 8. Analysis and Visualizations (Power BI)

**Dashboard Features:**

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**

  * **Multiple Visualizations based on problem statement:** Utilized diverse charts and visuals to communicate key insights effectively:

  * **Bar / Column Charts:** Used for Category-wise Sales Quantity and Freight Analysis, Product Performance by Total Quantity and Total Profit, Top Customers by Revenue, and Shipper Cost Comparison.

  * **Line & Area Charts:** Used for Sales_YTD by Month, Active_Customers and Sales_YTD by Date, and Forecast chart by Total sales, Sales YTD to track temporal and trend performance.

  * **Pie / Donut Charts:** Used for Sales Contribution (USA vs UK distribution), Employee Performance, and On_Time_Delivery_% gauge/donut visualization.

  * **Cards:** Used at the top of the dashboard to display key performance indicator (KPI) summary cards such as Total Sales ($1.27M), Total Profit ($251.24K), Sales YTD ($151.40K), Total Orders (388), Active Customers (87), and Total Freight Cost ($32.41K).

  * **Tables & Matrix Visuals:** Used for detailed data breakdowns, such as Profit_margin_% and Total_sales by Product Name and Active_vs_Discontinued_Category mapping.

- **Interactive Layout using Drill-down, Filters and slicers**

  * Enabled interactive data exploration by incorporating drill-through layers and hierarchical date filtering (Year, Quarter, Month, and Day).

  * Implemented dynamic report filters and slicers (such as Delivery_status [On Time / Delayed], Country [Argentina, Austria, Belgium, Brazil, Canada, etc.], and Product Discontinued flags) to let users slice and dice data seamlessly across pages

- **Bookmarks, Clear Titles for each visuals & Labels**

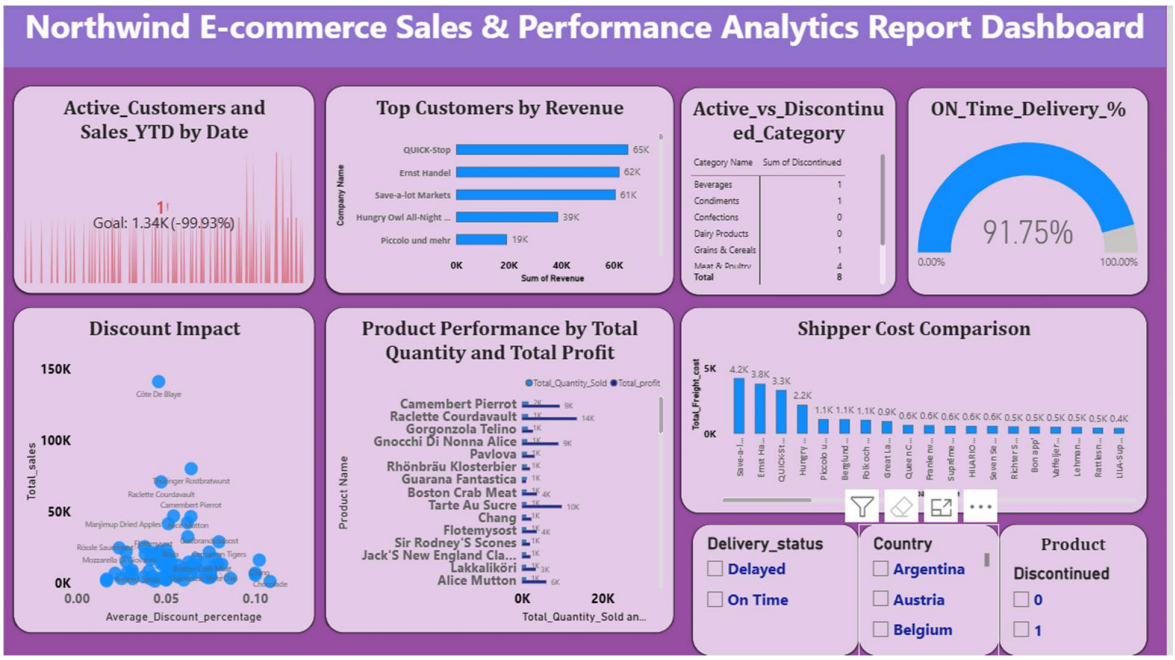
  * Configured synchronized buttons and navigation bookmarks to switch cleanly between different analytical views and report sections.

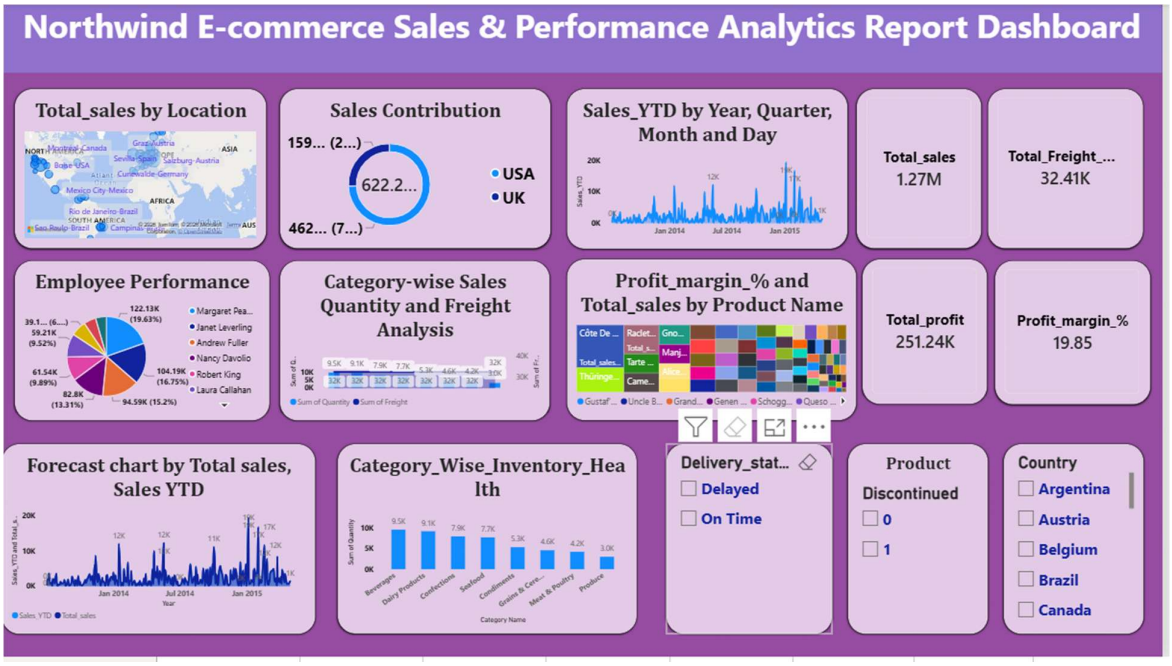
  * Applied clear, descriptive titles for every individual visual (e.g., Total_sales by Location, Discount Impact, Category_Wise_Inventory_Health) along with clear data labels and tooltips for enhanced readability.

- **Create a consolidated Report /Dashboard.**

  * Designed and integrated a Consistent, multi-page interactive dashboard titled "Northwind E-commerce Sales & Performance Analytics Report Dashboard" that combines sales performance, customer analytics, supply chain metrics, and inventory status into a unified executive view.

## SCREENSHOT OF DASHBOARD /REPORT



## 9. Insights & Conclusions

### ❖ Key Findings:

**Overall Revenue & Profitability:** The Northwind E-commerce platform successfully generated a total sales revenue of $1.27M with a net profit of $251.24K and an overall profit margin of 19.85%.

**Sales Growth & Time related Trends:** Year-to-date (YTD) sales reached $151.40K, demonstrating a positive year-over-year (YoY) growth rate of 4.59%. Monthly trends indicate peak sales activity during specific seasonal quarters.

**Geographical Distribution & Contribution**: Sales distribution shows strong concentration across international markets, supported by 87 active customers and 388 total orders.

**Supply Chain & Logistics Performance:** The on-time delivery rate stands at 91.75%, while total freight and shipping costs amounted to $32.41K across various shipping providers.

**Inventory & Workforce Impact:** Employee sales contributions totaled $622.20K, and category-wise analysis highlighted top-performing product lines alongside discontinued items affecting inventory health.

### Provide the analysis insights:

#### ❖ Descriptive

- **Total Sales & Profit:** The total sales for this e-commerce project reached 1.27M, with a total profit of 251.24K. This results in an overall profit margin of 19.85%.

- **Sales Contribution:** A major portion of the sales volume comes from key regions such as the USA and UK, with the USA alone contributing approximately 74.4% of total sales.

- **Customers & Orders:** Currently, the dataset reflects 388 total orders and 87 active customers. Additionally, the overall On-Time Delivery rate stands at 91.75%.

## ❖ Diagnosis

- **Category & Product Performance:** Categories such as 'Beverages' and 'Dairy Products' generate higher sales volumes and quantities sold. However, certain individual products show significantly lower profit margins compared to others.

- **Discount Impact:** The scatter plot analysis indicates that higher discount percentages (Average Discount Percentage: 0.06) increase sales quantity in some cases, but directly reduce the profit margin on specific items.

- **Shipper Cost Comparison**: Certain logistics/shipping partners incur higher freight costs, which serves as a major driver for the overall expenses (Total Cost: 1.01M).

## ❖ Predictive

- **Sales Forecasting:** Based on the forecast chart, sales performance and Sales YTD are expected to maintain a steady growth trend. However, periodic mid-month dips in sales volume indicate a need for targeted promotional efforts during those timeframes.

- **Customer Demand Trends:** Analysing purchasing patterns suggests that revenue from key

- Accounts will remain stable and continue to generate repeat orders.

## ❖ Prescriptive

- **Optimize Logistics Costs:** Review and renegotiate freight agreements with high-cost shipping partners, shifting more volume toward cost-effective shippers to minimize total logistics expenses.
- **Refine Discount Strategy:** Move away from flat discount rates and implement a targeted discount policy to safeguard profit margins on lower-margin products.

- **Enhance Customer Retention:** Introduce dedicated loyalty programs and tailored offers for top-tier customers to accelerate YoY sales growth (Sales YOY Growth: 4.59%).

## 10. Conclusions

The Northwind E-commerce Sales & Performance Analytics Dashboard provides a clear overview of overall financial health, product demand, and customer trends. While overall sales (1.27M) and profits (251.24K) demonstrate strong baseline performance, optimizing freight costs and refining discount strategies will further enhance net profit margins. Leveraging these descriptive, diagnostic, predictive, and prescriptive insights will enable data-driven business decisions for future growth.